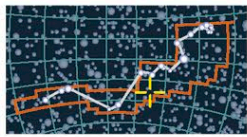


REGULAR CLUSTER

Hydra Cluster

CATALOG NUMBER Abell 1060

DISTANCE 160 million light-years



NUMBER OF GALAXIES 1,000+

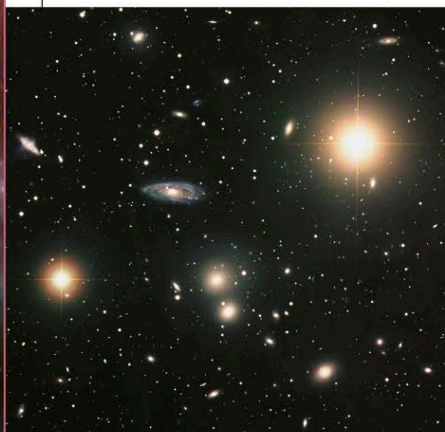
BRIGHTEST MEMBER NGC 3311 (11.6)

HYDRA

The Hydra Cluster is similar in size to the huge Virgo Cluster (see p.329). It is the closest example of a “relaxed” cluster (see p.327) of mainly elliptical galaxies in a spherical distribution. Its hot X-ray gas also forms a spherical cloud around the core. The cluster is centered on two giant elliptical galaxies and an edge-on spiral, each 150,000 light-years across. These galaxies are interacting—the ellipticals’ gravity has warped the spiral, while both ellipticals have distorted outer haloes. The cluster is the major member of the Hydra Supercluster, which adjoins the Local Supercluster (see pp.336–339).

HEART OF THE HYDRA CLUSTER

In this image, the central giant ellipticals NGC 3309 and 3311 lie below the large, blue spiral NGC 3312. The two bright objects on either side are foreground stars.



SPIRAL SILHOUETTE

NGC 3314, an unusual case of one spiral galaxy silhouetted against another, is one of Hydra’s most beautiful objects.



COMPACT GROUP

Stephan’s Quintet

CATALOG NUMBER

Hickson 92

DISTANCE

340 million light-years (NGC 7320: 41 million light-years)

NUMBER OF GALAXIES 4/5

BRIGHTEST MEMBER NGC 7320 (13.6)



PEGASUS

First observed by French astronomer E. M. Stephan at the University of Marseilles in 1877, Stephan’s Quintet appears to be a remarkably compact cluster of five galaxies. The galaxies are a mixture of spirals, barred spirals, and ellipticals and show clear signs of disruption from interactions. The largest galaxy as seen from Earth, NGC 7320, is probably a foreground object lying in front of a quartet of interacting galaxies. The spectral red shift (see p.35) of NGC 7320 is much smaller than those of the other four galaxies and instead matches that of several other galaxies close to it in the sky. Because it also appears physically different from the quartet, it seems likely that NGC 7320 is much closer and the unusual red shift is a normal result of the expansion of space (see p.44). However, a few astronomers claim that trails of material link NGC 7320 to other Quintet galaxies. If this is the case, then the red shift suggests that the galaxy is moving very fast relative to its neighbors and toward Earth, therefore reducing its overall speed of recession and its red shift. Or perhaps the red shift does not originate from its motion at all. These competing theories have turned Stephan’s Quintet into a battleground for the small minority of astronomers who think that red shifts are not all caused by the expansion of space and that Hubble’s Law (see p.44) does not always apply.

FOUR OR FIVE?

The quintet consists of a quartet of yellow galaxies beside the white spiral NGC 7320. The contrasting appearance of NGC 7320 suggests it lies in front of the other galaxies.



QUINTET CLOSE-UP

This detailed Hubble Space Telescope view of Stephan’s Quintet shows chains of stars linking several of its interacting galaxies.



REGULAR CLUSTER

Coma Cluster

CATALOG NUMBER Abell 1656

DISTANCE 300 million light-years

NUMBER OF GALAXIES 3,000+

BRIGHTEST MEMBER NGC 4889 (13.2)



COMA BERENICES

Although it lies near the Virgo Cluster in the sky (see p.329), the Coma Cluster is much farther away. First recognized by William Herschel as a concentration of “fine nebulae”

in 1785, this is one of the nearest highly evolved or “relaxed” galaxy clusters (see p.327). It is very dense, with over 3,000 galaxies, and is dominated by elliptical and lenticular galaxies. Because it is near the north galactic pole (and therefore free of the dense star fields of the Milky Way), it is well studied. Swiss-American astronomer Fritz Zwicky used Coma when he made the first measurements of galaxy movements within a cluster in the 1930s. He found the cluster contained many times more mass than its visible galaxies suggested—an idea that was not accepted until the 1970s. Overall, the cluster is moving away at 16 million mph (25 million kph). At

the cluster’s center lie the giant elliptical NGC 4889 and the lenticular galaxy NGC 4874. Most of the spirals and irregulars are in the outer regions. X-ray images show two distinct patches of cluster gas, suggesting that the cluster is absorbing a smaller cluster of galaxies. Like the Virgo and Hydra Clusters, Coma forms the core of its own galaxy supercluster.

HIDDEN DWARFS

This combined visible and infrared image reveals dozens of faint green smudges, marking the location of dwarf galaxies that are too faint to see in visible light.

